Rankers Test Series (2023)

PART TEST–09

NEET

DURATION : 200 Minutes | **12-03-2023** | **M. MARKS : 720**

Topics Covered

Physics :	Electromagnetic Induction, Alternating Current, Electromagnetic Waves
Chemistry :	The p-Block Elements, The d and f Block elements, Coordination Compounds, Haloalkanes and Haloarenes, Alcohols
Biology :	**(Botany) :** Molecular Basis of Inheritance **(Zoology) :** Evolution

General Instructions:

1. Immediately fill in the particulars on this page of the test booklet.
2. The test is of **3 hour 20 minute** duration.
3. The test booklet consists of **200** questions. The maximum marks are **720**.
4. There are four Section in the Question Paper, Section I, II, III & IV consisting of Section-I **(Physics)**, Section-II **(Chemistry)**, Section-III **(Botany)** & Section IV **(Zoology)** and having **50 Questions** in each part.
5. There is only one correct response for each questions.
6. Each correct answer will give 4 marks while 1 Mark will be deducted for a wrong MCQ response.
7. No student is allowed to carry any textual material, printed or written, bits of papers, pager, mobile phone, any electronic device, etc. inside the examination room/hall.
8. On completion of the test, the candidate must hand over the Answer Sheet to the Invigilator on duty in the Room/Hall. However, the candidates are allowed to take away this Test Booklet with them.

OMR Instructions:

1. Use blue/black dark ballpoint pens.
2. Darken the bubbles completely. Don't put a tick mark or a cross mark where it is specified that you fill the bubbles completely. Half-filled or over-filled bubbles will not be read by the software.
3. Never use pencils to mark your answers.
4. Never use whiteners to rectify filling errors as they may disrupt the scanning and evaluation process.
5. Writing on the OMR Sheet is permitted on the specified area only and even small marks other than the specified area may create problems during the evaluation.
6. Multiple markings will be treated as invalid responses.
7. **Do not fold or make any stray mark on the Answer Sheet (OMR).**

Name of the Student (In CAPITALS) : _____

Roll Number : _____

OMR Bar Code Number : _____

Candidate's Signature : _____ **Invigilator's Signature** _____

SECTION-I (PHYSICS)

SECTION: A

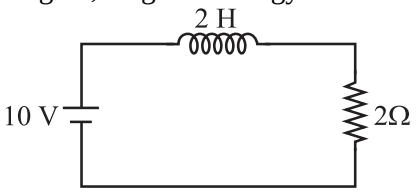
1. The magnetic flux across a loop of resistance 10Ω is given by $\phi = (5t^2 - 4t + 1)$ Wb. How much current is induced in the loop just after 0.2 sec?
(1) 0.4 A
(2) 0.2 A
(3) 0.04 A
(4) 0.02 A

2. A 0.1 m long conductor carrying a current of 50 A is held perpendicular to a magnetic field of 1.25 mT. The mechanical power required to move the conductor with a speed of 1 m sec^{-1} is
(1) 62.5 mW
(2) 625 mW
(3) 6.25 mW
(4) 12.5 mW

3. In circular coil, when number of turns is doubled and resistance becomes $\frac{1}{4}$ th of initial, then inductance becomes–
(1) 4 times (2) 2 times
(3) 8 times (4) No charge

4. A current is varying at the rate of 3 A/sec in a coil generates an emf of 8 mV in a nearly coil. The mutual inductance of the two coils is
(1) 2.66 mH
(2) 2.66×10^{-3} mH
(3) 2.66 H
(4) 0.266 H

5. In the figure, magnetic energy stored in the coil is

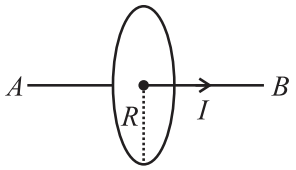


(1) 0 (2) ∞
(3) 25 J (4) None of these

6. The time taken for the magnetic energy to reach 25% of its maximum value, when a solenoid of resistance R and inductance L is connected to a battery, is–
(1) $\frac{L}{R}(\ln 2)$
(2) $\frac{L}{R}(\ln 10)$
(3) Infinite
(4) $\frac{L}{R}(\ln 5)$

7. An inductance L and a resistance R are first connected to a battery. After some time the battery is disconnected but L and R remain connected in a close circuit. Then the current reduces to 37% of its initial value in time equal to
(1) RL (2) $\frac{R}{L}$
(3) $\frac{L}{R}$ (4) $\frac{1}{LR}$

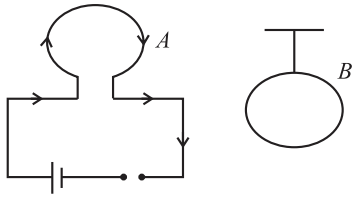
8. A conductor AB lies along the axis of a circular loop of radius R. If the current in the conductor AB varies at the rate of I ampere/sec, the induced emf in the loop is:



(1) $\frac{\mu_0 IR}{2}$ (2) $\frac{\mu_0 IR}{4}$
(3) $\frac{\mu_0 \pi IR}{2}$ (4) Zero

9. **Assertion:** If two inductors are in parallel, then current in them distributes in inverse ratio of their inductances.
Reason: In parallel, potential difference remains constant.
(1) If both Assertion and Reason are true and Reason is the correct explanation of Assertion.
(2) If both Assertion and Reason are true but Reason is not the correct explanation of Assertion.
(3) If Assertion is true but Reason is false.
(4) If both Assertion and Reason are false.

10. A system S consists of two coils A and B. The coil A carries a steady current I while the coil B is suspended nearly as shown in figure. Now, if the system is heated, so as to raise the temperature of two coils steadily, then



(1) The two coils show attraction
(2) The two coils show repulsion
(3) There is no change in the position of the two coils
(4) Induced current is not possible in coil B

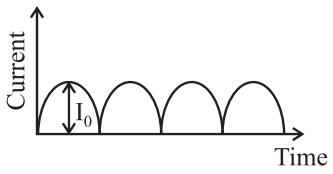
11. An *L*-*R* circuit has $R = 10\Omega$ and $L = 2$H. If 120 V, 60 Hz AC voltage is applied, then current in the circuit will be

(1) 0.32 A (2) 0.16 A
(3) 0.48 A (4) 0.80 A

12. In a series resonant *LCR* circuit, the voltage across *R* is 100 volts and $R = 1\Omega$ with $C = 2\mu F$. The resonant frequency ω is 200 rad/sec. At resonance, the voltage across *L* is:

(1) 4×10^{-3} volt (2) 2.5×10^{-2} volt
(3) 40 volt (4) 2.5×10^{5} volt

13. The output sinusoidal current versus time curve of a rectifier is shown in the figure. The average value of output current in this case is

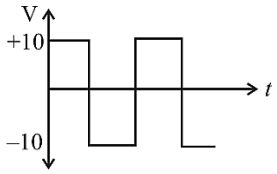


(1) 0 (2) $\frac{I_0}{2}$
(3) $\frac{2I_0}{\pi}$ (4) I_0

14. In a transformer, numbers of turns in the primary are 140 and that in the secondary are 280. If current in primary is 4A, then in the secondary is

(1) 4A (2) 2A
(3) 6A (4) 10A

15. The rms voltage of the waveform shown is

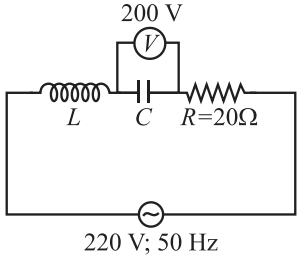


(1) 10 V (2) 7 V
(3) 6.37 V (4) None of these

16. In a series *L*-*R*-*C* circuit resistance $R = 10\Omega$ and the impedance $Z = 10\Omega$. The phase difference between the current and the voltage is

(1) 0° (2) 30°
(3) 45° (4) 60°

17. In the circuit shown, rms current is 11A. The potential difference across the inductor is

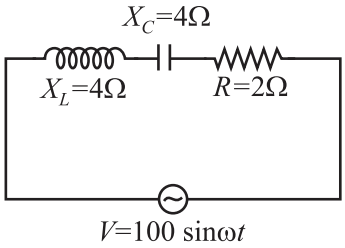


(1) 220 V (2) 0 V
(3) 300 V (4) 200 V

18. An *L*-*C* circuit contains a 20 mH inductor and 50 μF capacitor with an initial charge of 10 mC. The resistance of the circuit is negligible. At the instant $t = 0$ the circuit is closed. At what time is the energy stored completely magnetic?

(1) $t = 0$ (2) $t = 1.57$ ms
(3) $t = 3.14$ ms (4) $t = 6.28$ ms

19. Which of the following statement is correct regarding the AC circuit shown in the adjacent figure?



(1) The rms value of current through the circuit is $i_{rms} = 5\sqrt{2}$ A
(2) The phase difference between source emf and current is $\phi = \cos^{-1}\left(\frac{1}{3}\right)$
(3) Average power dissipated in the circuit is 500 W
(4) None of the above

20. A condenser of 250 pF is connected in parallel to a coil of inductance of 0.16 mH, while its effective resistance is 20Ω. Determine the resonant frequency.

(1) 9×10^4 Hz (2) 16×10^7 Hz
(3) 8×10^5 Hz (4) 9×10^3 Hz

21. The average electric field of electromagnetic waves in certain region of free space is 9×10^{-4} NC^{-1}. Then the average magnetic field in the same region is of the order of

(1) 27×10^{-4} T (2) 3×10^{-12} T
(3) $\left(\frac{1}{3}\right) \times 10^{-12}$ T (4) 3×10^{12} T

22. A parallel plate capacitor is charged to 60 μC. Due to a radioactive source, the plate loses charge at the rate of 1.8×10^{-8} $Csec^{-1}$. The magnitude of displacement current is

(1) 1.8×10^{-8} C/sec
(2) 3.6×10^{-8} C/sec
(3) 4.1×10^{-11} C/sec
(4) 5.7×10^{-12} C/sec

23. The electric field intensity produced by the radiations coming from 100W bulb at a 3 m distance is *E*. The electric field intensity produced by the radiations coming from 50 W bulb at the same distance is

(1) $\frac{E}{2}$ (2) $2E$
(3) $\frac{E}{\sqrt{2}}$ (4) $\sqrt{2}E$

24. **Assertion:** Ultraviolet radiations of higher frequency wave are dangerous to human being.
Reason: Ultraviolet radiation are absorbed by the atmosphere.
(1) If both Assertion and Reason are true and Reason is the correct explanation of Assertion.
(2) If both Assertion and Reason are true but Reason is not the correct explanation of Assertion.
(3) If Assertion is true but Reason is false.
(4) If both Assertion and Reason are false.

25. In an electromagnetic wave, the electric and magnetic fields are 100 V/m and 0.265 T. The maximum energy flow is:
(1) 26.5 W/m^2 (2) 36.5 W/m^2
(3) 46.7 W/m^2 (4) 765 W/m^2

26. Radiations of intensity 0.5 W/m^2 are striking a metal plate. The pressure on the plate is
(1) 0.166×10^{-8} N/m^2
(2) 0.322×10^{-8} N/m^2
(3) 0.111×10^{-8} N/m^2
(4) 0.083×10^{-8} N/m^2

27. Light with an energy flux of 20 W/cm^2 falls in a non-reflecting surface at normal incidence. The surface has an area of 30 cm^2. The total momentum delivered (for complete absorption) during 30 minutes is
(1) 36×10^{-5} kg-m/sec
(2) 36×10^{-4} kg-m/sec
(3) 1.08×10^{4} kg-m/sec
(4) 1.08×10^{7} kg-m/sec

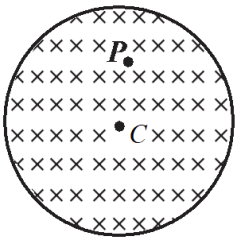
28. The charge on a parallel plate capacitor varies as $q = q_0 \cos 2\pi ft$. The plates are very large and close together (area = A, separation = d). Neglecting the edge effects, find the displacement current through the capacitor.
(1) $-4q_0\pi f \sin(2\pi ft)$
(2) $-2q_0\pi f \sin(2\pi ft)$
(3) $-3q_0\pi f \sin(2\pi ft)$
(4) $q_0\pi f \sin(2\pi ft)$

29. A radio can tune into any station in the 7.5 MHz to 12 MHz band. What is the corresponding wave length band?
(1) 80 m to 40 m (2) 85 m to 40 m
(3) 45 m to 50 m (4) 40 m to 25 m

30. What physical quantity is the same for X-ray of wavelength 10^{-10} m, red light of wavelength 6800 Å and radio waves of wavelength 500 m?
(1) Frequency (2) Speed
(3) Distance (4) None of these

31. An emf of 15 V is applied in a circuit coil containing 5 H inductance and 10 Ω resistance. The ratio of the currents at time $t = \infty$ and $t = 1$ sec is
(1) $\dfrac{e^{\frac{1}{2}}}{e^{\frac{1}{2}}-1}$
(2) $\dfrac{e^{2}}{e^{2}-1}$
(3) $1 - e^{-1}$
(4) e^{-1}

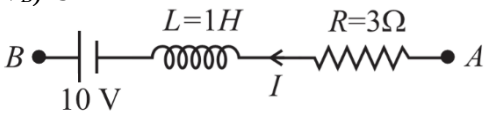
32. A uniform but increasing with time magnetic field exists in a cylindrical region. The direction of force on an electron at P is



(1) Towards right
(2) Towards left
(3) Into the plane of paper
(4) Out of the plane of paper

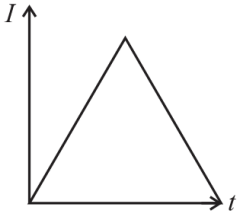
33. Magnetic flux during time interval τ varies through a stationary loop of resistance R as $\phi_B = at(\tau - t)$. Find the amount of heat generated during that time. Neglect the inductance of the loop.
(1) $\dfrac{a^2\tau^3}{R}$
(2) $\dfrac{a^2\tau^2}{2R}$
(3) $\dfrac{a^2\tau^3}{3R}$
(4) $\dfrac{a^2\tau^3}{4R}$

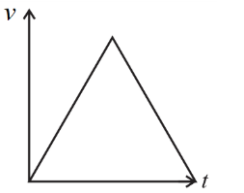
34. In the given branch AB of a circuit a current, $I = (10t + 5)A$ is flowing, where t is time in second. At $t = 0$, the potential difference between point A and B ($V_A - V_B$) is–

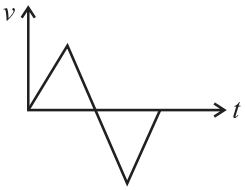


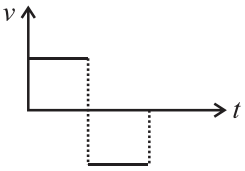
(1) 15 V
(2) – 5 V
(3) – 15 V
(4) 5 V

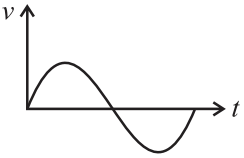
35. An alternating current I in an inductance coil varies with time t according to the graph as shown which one of the following graphs gives the variation of voltage with time?



(1)


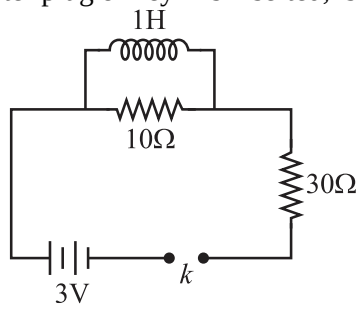
(2)


(3)


(4)


SECTION – B

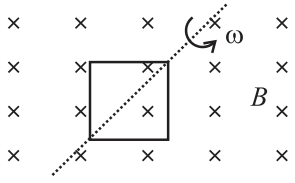
36. In figure, final value of current in 10Ω resister, long time after plug of key k is inserted, is



(1) $\frac{3}{10}$A (2) $\frac{3}{20}$A

(3) $\frac{3}{11}$A (4) Zero

37. A square loop of side b is rotated in a constant magnetic field B at angular frequency ω as shown in the figure. What is the emf induced in it?



(1) $b^2B\omega \sin(\omega t)$ (2) $bB\omega \sin(\omega t)$

(3) $bB^2\omega \cos(\omega t)$ (4) $b^2B\omega$

38. Voltage applied to an AC circuit and current flowing in it is given by $V = 200\sqrt{2}\sin\left(\omega t + \frac{\pi}{4}\right)$ and $i = -\sqrt{2}\cos\left(\omega t + \frac{\pi}{4}\right)$. Then, power consumed in the circuit will be

(1) 200 W (2) 400 W

(3) $200\sqrt{2}$ W (4) None of these

39. The frequency of an alternating current is 50 Hz. The minimum time taken by it in reaching from zero to peak value is

(1) 5 ms (2) 10 ms

(3) 20 ms (4) 50 ms

40. If L and R be the inductance and resistance of the choke coil, then identify the correct statement.

(1) L is very high compared to R.

(2) R is very high compared to L.

(3) Both L and R are high.

(4) Both L and R are low.

41. In an AC circuit, the impedance is $\sqrt{3}$ times the reactance, then the phase angle is

(1) 60° (2) 30°

(3) Zero (4) None of these

42. A complex current wave is given by $i = (5 + 5 \sin 100\omega t)$A. Its average value over one time period is given as

(1) 10 A (2) 5 A

(3) $\sqrt{50}$ A (4) 0

43. The magnetic field of an electromagnetic wave is given by, $B_y = 3 \times 10^{-7} \sin(10^3 x + 6.28 \times 10^{12} t)$. The wavelength of the electromagnetic wave is:

(1) 6.28 mm (2) 0.63 mm

(3) 3.14 mm (4) 0.32 mm

44. In an electromagnetic wave, the amplitude of electric field is 1 V/m. The frequency wave is 5×10^4 Hz. The wave is propagating along z-axis. The average energy density of electric field in joule/m^3, will be

(1) 1.1×10^{-11} (2) 2.2×10^{-12}

(3) 3.3×10^{-13} (4) 4.4×10^{-14}

45. The self-inductance of two coils of a transformer are 20 mH and 30 mH. What is the resistance between them?

(1) Infinite

(2) Zero

(3) 1.5Ω

(4) 600Ω

46. The magnetic flux linked with a circuit of resistance 100Ω increases from 10 to 60 Wb. The amount of induced charge that flows in the circuit is (in coulomb)

(1) 0.5 (2) 5

(3) 50 (4) 100

47. If emf induced in a coil is 2V by changing the current in it from 8 A to 6 A in 2×10^{-3} sec, then the coefficient of self-induction is

(1) 2×10^{-3} H (2) 10^{-3} H

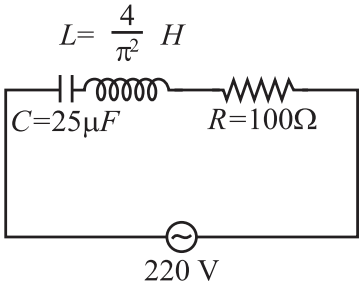
(3) 0.5×10^{-3} H (4) 4×10^{-3} H

48. The average power dissipated in AC circuit is 2 W. If a current flowing through a circuit is 2A, impedance is 1Ω, then what is the power factor of the circuit?

(1) 0.5 (2) 1

(3) Zero (4) $\frac{1}{\sqrt{2}}$

49. The resonant frequency of the given *RLC* circuit.



(1) 50 Hz (2) 100 Hz

(3) 150 Hz (4) 200 Hz

50. An alternating emf 200 volts at 50 Hz is connected to a circuit of resistance 1Ω and inductance 0.01 H. What is the phase difference between the current and the emf in the circuit?

(1) $\tan^{-1}(2\pi)$ (2) $\tan^{-1}(\pi)$

(3) $\tan^{-1}\left(\frac{\pi}{2}\right)$ (4) $\tan^{-1}\left(\frac{3\pi}{2}\right)$

SECTION-II (CHEMISTRY)

SECTION-A

51. Dinitrogen exists as a diatomic linear molecule but phosphorous exists as tetra atomic tetrahedral molecule. Why?

(1) Tetrahedral structure is more stable than linear.

(2) Phosphorous cannot form pπ – pπ bond as the length of P–P σ-bond is very large due to which p-orbitals are unable to overlap to give π-bond

(3) Phosphorous cannot form hydrogen bond

(4) Size of P is very small

52. Except _____, all the member of N-family forms allotropes.

(1) N (2) P

(3) As (4) Bi

53. Out of all the member of N-family, only _____ does not form complex by accepting electron pairs.

(1) N (2) P

(3) As (4) Sb

54. The bond angle of hydrides of elements of Chalcogens decreases down the group because

(1) The size and electronegativity of central atom increases down the group

(2) The size increases and electronegativity decrease of central atom down the group

(3) The size decreases and electronegativity increase of central atom down the group

(4) The size and electronegativity of central atom decreases down the group

55. Which of the following is correct?

(1) Pure ozone is a pale blue gas, dark-blue liquid and violet-black solid.

(2) Pure ozone is a pale-pink gas, dark-red liquid and brown solid.

(3) Pure ozone is a pale-yellow gas, dark-yellow liquid and orange solid.

(4) Pure ozone is a pale grey gas, dark-grey liquid and silvery solid.

56. A dioxide of sulphur reacts with molecular halogen gas in the presence of charcoal to give a liquid A which on oxidation using molecular oxygen in the presence of vanadium(V) oxide to give a trioxide B of an element having mass number 32. Identify A and B.

(1) SO_3 and SO_2Cl_2 (2) SO_3 and $SOCl_2$

(3) SO_2Cl_2 and SO_3 (4) $SOCl_2$ and SO_3

57. Match the following

Column I	Column II
A. Oleum	1. $H_2S_2O_8$
B. Caro's acid	2. H_2SO_5
C. Marshall's Acid	3. $H_2S_2O_7$

	A	B	C
(1)	1	2	3
(2)	3	2	1
(3)	2	3	1
(4)	3	1	2

58. **Assertion:** Contact process is an exothermic, reversible and the forward reaction leads to decrease in volume.
Reason: For Maximum yield, the reaction is operated at low temperature and high pressure.
(1) Assertion is correct, reason is correct; reason is a correct explanation for assertion.
(2) Assertion is correct, reason is correct; reason is not a correct explanation for assertion
(3) Assertion is correct, reason is incorrect
(4) Assertion is incorrect, reason is correct

59. Name the hydrogen halide having the longest bond length.
(1) HI (2) HBr
(3) HCl (4) HF

60. If we take hypochlorous acid, chlorous acid, chloric acid and perchloric acid, which is the strongest acid?
(1) Hypochlorous acid
(2) Chlorous acid
(3) Chloric acid
(4) Perchloric acid

61. Which among the following has low melting and boiling point?
(1) Copper (2) Nickel
(3) Iron (4) Zinc

62. The maximum number of oxidation states are shown by
(1) Cr
(2) Mn
(3) Fe
(4) Both (1) and (2)

63. Among the elements of first transition series, the element having the most negative reduction potential for M^{+3}/M system.
(1) Scandium
(2) Manganese
(3) Titanium
(4) Vanadium

64. The first series of transition elements have negative standard reduction potential except, copper. Therefore, they should be good reducing agents but they are weaker reducing agents in comparison to group 1, 2, 13 elements. Why?
(1) High heats of vaporisation
(2) High ionisation enthalpies
(3) Low heat of hydration of their ions.
(4) All of the above

65. The effective magnetic moment of transition metal containing unpaired electron is the magnetic moment due to spin angular moment of the electron. The orbital magnetic moment is not considered due to
(1) Quenching of the orbital magnetic moment by the surrounding ligands.
(2) d- orbitals are seated deeply into the atom.
(3) f-orbitals are seated outside the atom
(4) None of the above

66. The reason behind the catalytic property of d-block element is
(1) High ionization enthalpy
(2) Low heat of hydration
(3) Negative standard reduction potential
(4) Variable valency

67. **Assertion:** Transition elements form complexes
Reason: The transition metals ions have very small size and have high positive charge density. Also, they have vacant orbitals having right type and amount of energy to accept lone pair.
(1) Assertion is correct, reason is correct; reason is a correct explanation for assertion.
(2) Assertion is correct, reason is correct; reason is not a correct explanation for assertion
(3) Assertion is correct, reason is incorrect
(4) Assertion is incorrect, reason is correct

68. Among 15 lanthanoids, the metals which can only have +3 oxidation state.
(1) La, Ce, Pr (2) Nd, Pm, Sm
(3) Er, Tm, Yb (4) La, Gd, Lu

69. Which of the following is false?
(1) Lanthanoid hydroxide is ionic and basic in nature.
(2) Lanthanoid hydroxides are stronger base than aluminium hydroxide.
(3) Lanthanoid hydroxides are weaker base than calcium hydroxide.
(4) As we move from La to Lu hydroxides, the basicity increases.

70. Which of the following statement are correct?
A. Lanthanoid ions have low charge density as they have large size.
B. They have low tendency to form complexes especially with unidentate ligand.
C. They preferably form chelating complexes.
D. They form π-acid complexes.
(1) ABC (2) BCD
(3) CDA (4) None

71. If a solution containing mixture of tri positive lanthanoid ions of metals Lu, La, Gd, Tm made to pass through ion-exchange chromatography under the wash of effective eluent, the lanthanoid which comes out first is
(1) Lu (2) La
(3) Gd (4) Tm

72. Which of the following is the tridentate ligand?
(1) Py (2) Dipy
(3) Terpy (4) Trien

73. When a complex $PtCl_4 \cdot 2NH_3$ is treated with excess of silver nitrate solution, no white ppt is formed. The expected structure of the given complex is
(1) $[PtCl_2(NH_3)_2]Cl_2$ (2) $[PtCl_4(NH_3)_2]$
(3) $[PtCl_3(NH_3)_2]Cl$ (4) $[PtCl(NH_3)_2]Cl_3$

74. When $PtCl_4 \cdot 6NH_3$ is treated with excess of silver nitrate solution, four moles of silver chloride precipitated out. The total number of ions for the complex is
(1) 5 (2) 4
(3) 3 (4) 6

75. Which of the following is not an ambidentate ligand?
(1) Nitro (2) Thiocyanato
(3) Cyanato (4) Chloro

76. The stability constant of a complex is 10^{-3}. The dissociation constant of the complex is
(1) 10^3 (2) 10^6
(3) 10^9 (4) 10^{-6}

77. From the Irving William series, the log K value of complex formation of ferrous ion is more than that of calcium ion because
(1) Iron is transition metal
(2) Size is small
(3) Octahedral complex of iron is additionally stabilised by CFSE
(4) High charge to size ratio

78. **Assertion:** The octahedral complex of cupric ion is more stable than that of nickel ion.
Reason: The CFSE of former is less than that of latter but additional stability is given by Jahn-Teller Distortion.
(1) Assertion is correct, reason is correct; reason is a correct explanation for assertion.
(2) Assertion is correct, reason is correct; reason is not a correct explanation for assertion
(3) Assertion is correct, reason is incorrect
(4) Assertion is incorrect, reason is correct

79. The isomerism shown by $CrCl_3 \cdot 6H_2O$ complex is
(1) Ionisation (2) Hydrate
(3) Coordination (4) Linkage

80. The isomerism shown by
$[Pt(NO_2)(C_5H_5N)(NH_3)(NH_2OH)]^+$ is
(1) Ionisation (2) Geometrical
(3) Optical (4) Coordination position

81. If you are given with d^4 system, the CFSE for high spin and low spin octahedral complex would be (in Dq)
(1) 6 & 16 (2) 16 & 6
(3) 6 & 6 (4) 16 & 16

82. Which of the following is true?
(1) CFSE depends only on the nature of the ligands
(2) Big size ligands can approach the metal very easily and cause high CFSE
(3) CO is a strong field ligand than ammonia.
(4) The metal with higher oxidation state cause less CFSE value.

83. The nature of bonding in metal carbonyls is
(1) Purely σ donation
(2) Purely π donation
(3) σ donation and π acceptation
(4) π donation and σ acceptation

84. The hybridisation caused in $[FeF_6]^{3-}$ complex is
(1) sp^3d^2 and outer sphere complex
(2) sp^3d^2 and inner sphere complex
(3) d^2sp^3 and outer sphere complex
(4) d^2sp^3 and inner sphere complex

85. The d-orbital involved in pentacarbonyl iron (0) complex is
(1) d_{xy} (2) d_{yz}
(3) d_{xz} (4) d_{z^2}

SECTION B

86. Arrange the following alkyl bromides in order of decreasing reactivity in an S_N2 reaction:
1-bromo-2-methylbutane(A),
1-bromo-3-methylbutane(B),
2-bromo-2-methylbutane(C),
and 1-bromopentane(D).
(1) C < A < B < D
(2) A < C < D < B
(3) D < A < B < C
(4) B < C < D < A

87. How will you distinguish between methyl alcohol and ethyl alcohol?
(1) Lucas test (2) Iodoform test
(3) Carbylamine test (4) None

88. Arrange the following compounds, which are nearly of identical molecular weights, in order of increasing boiling points.
A. 1-butanamine
B. 1-butanol
C. Methyl n-propyl ether
D. Pentane
(1) D < C < A < B (2) A < B < C < D
(3) D < C < B < A (4) C < B < D < A

89. Arrange the following compounds in order of increasing solubility in water.
A. Ethanol
B. Ethyl chloride
C. 1-hexanol
(1) A < B < C (2) B < C < A
(3) C < A < B (4) A < C < B

90. Which combination is known as the "Lucas Reagent"?
(1) NaCl, HCl, $CuCl_2$
(2) HCl, $ZnCl_2$
(3) HCl, $SnCl_2$
(4) NaCl, $SnCl_2$

91. For which of the following oxidation reaction does one need to use PCC?
(1) Methanol $\rightarrow$ Formic acid
(2) Ethanol $\rightarrow$ Acetic acid
(3) Ethanol $\rightarrow$ Acetaldehyde
(4) Acetaldehyde $\rightarrow$ Acetic acid

92. Which compounds can be reduced easily with sodium borohydride?
(1) Amide
(2) Esters
(3) Aldehyde and ketones
(4) Alkenes

93. What role does pyridine play in PCC reagent?
(1) To make chromium reagent less acidic.
(2) To make chromium reagent a stronger oxidizing agent
(3) To make the chromium compound water soluble
(4) To make chromium reagent soluble in nonpolar organic solvents

94. In the E_1 reaction, how is the rate affected when the concentration of attacking nucleophile is doubled?
(1) Doubled
(2) Halved
(3) Quadruples
(4) Unchanged

95. In the reaction of 1-chloropentane and potassium tert-butoxide, the dominant mechanism is
(1) S_N1 (2) S_N2
(3) E_1 (4) E_2

96. Which is the most basic nucleophile?
(1) Fluoride (2) Ethanol
(3) Water (4) Ethoxide

97. Which of the following is a true statement?
(1) The weaker the base, the better it is as a leaving group.
(2) The smaller the group, the better it is as a leaving group.
(3) The stronger the base, the better it is as a leaving group
(4) The more electronegative the group, the better it is as a leaving group

98. What product is formed when para-methyl phenol reacts with Br_2 with $FeCl_3$?
(1) Mainly 3-bromo-4-methyl phenol
(2) Mainly 2-bromo-4-methyl phenol
(3) 2-bromo-4-methyl phenol and 3-bromo-4-methyl phenol in comparable amount
(4) Mainly 4-bromomethyl phenol

99. What function does nitric acid serve in the reaction of benzene with iodine to produce Iodobenzene?
(1) The acid oxidizes iodine to iodine cation
(2) The acid converts iodine to iodide anion
(3) The acid converts iodine to hydrogen iodide acid.
(4) The acid serves as a catalyst.

100. Which of the following is not an ortho-para director?
(1) – Cl
(2) – Br
(3) – CN
(4) – F

SECTION-III (BOTANY)

SECTION-A

101. DNA was first identified by
(1) Miescher (2) Watson
(3) Griffith (4) Francis Crick

102. The process of copying genetic information from one strand of DNA into RNA is
(1) Translation (2) Replication
(3) Transformation (4) Transcription

103. DNA replication takes place in which phase of cell cycle?
(1) G_1 phase (2) G_2 phase
(3) S-phase (4) M-phase

104. In Meselson and Stahl experiment, heavy DNA distinguished from normal DNA by centrifugation in which density gradient medium?
(1) $^{14}NH_4Cl$ (2) $^{15}NH_4Cl$
(3) CsCl (4) Both (1) and (2)

105. Taylor and his colleagues performed experiments on *vicia faba* using
(1) Radioactive cytidine
(2) Radioactive thymidine
(3) Radioactive uridine
(4) Radioactive sulphur

106. The correct formula of Chargaff rule is
(1) A + G = T + C
(2) A + T = G + C
(3) A + C = G + T
(4) All are incorrect

107. The percentage of guanine is 18 in a double stranded DNA molecule, what would be the percentage of thymine in it?
(1) 18 (2) 36
(3) 62 (4) 32

108. The unequivocal proof that DNA is the genetic material given by which experiment and performed using _____?
(1) Hershey – chase using bacterial virus
(2) Transforming principle using bacteria
(3) Meselson and Stahl using bacteria
(4) Biochemical method using enzymes

109. Length of DNA is affected by
(1) Nitrogenous bases
(2) Sugar
(3) Pitch
(4) Phosphate

110. Histones are rich in
(1) Lysine (2) Arginine
(3) Serine (4) Both (1) and (2)

111. In a double stranded DNA
(1) There is unequal proportion of purines and pyrimidine
(2) Pitch of the helix is 3.4 nm
(3) The two strand run parallel in 5′ → 3′ direction
(4) Guanine is bonded with thymine by three hydrogen bond

112. Read the following statements and choose the correct option
(a) Two Nucleotides are linked through phosphoester bond.
(b) Nitrogen base is linked to pentose sugar by N-glycosidic linkage.
(c) Thymine is also referred to as 5-ethyl-uracil.
(d) Presence of thymine confers additional stability to DNA.
(e) Nucleotides are formed by phosphodiester bond.
(1) a, b, c, e are correct
(2) b, c, d are correct
(3) b, d are correct
(4) All are correct

113. A typical nucleosome contains _____ bp of DNA helix.
(1) 200 base pairs
(2) 100 base pairs
(3) 300 base pairs
(4) 3.3×10^9 base pairs

114. Match the column I with suitable terms in column II

Column I		Column II	
A.	Bacteriophages	i.	mRNA processing
B.	Capsid	ii.	Transposons
C.	Splicing	iii.	Protein coat
D.	Okazaki fragments	iv.	Viruses
E.	DNA with switching position	v.	DNA

	A	B	C	D	E
(1)	iv	iii	v	ii	i
(2)	iv	iii	i	v	ii
(3)	v	ii	iii	i	iv
(4)	iii	v	i	ii	iv

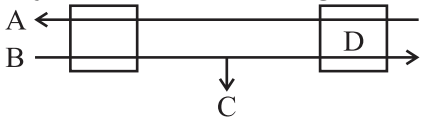
115. RNA polymerase I transcribes which of the following RNA?
(1) rRNA (2) tRNA
(3) SnRNA (4) mRNA

116. How many codons code for different types of amino acids?
(1) 32 (2) 61
(3) 64 (4) 60

117. RNA polymerase binds to which of the following factors to terminate transcription?
(1) Gamma (2) Alpha
(3) Sigma (4) Rho

118. **Statement I:** RNA polymerase are present only in nucleus.
Statement II: RNA polymerase catalyse only process of elongation.
(1) Both statements are correct
(2) Both statements are incorrect
(3) Only statement I is correct
(4) Only statement II is correct

119. Identify A, B, C and D for the given diagram?



(1) A → 3′ → 5′, B → 5′ → 3′, C → Coding strand, D → Terminator
(2) A → 5′ → 3′, B → 3′ → 5′, C → Template strand, D → Terminator
(3) A → 3′ → 5′, B → 5′ → 3′, C → Template strand, D → Terminator
(4) All are incorrect

120. RNA polymerase III transcribes
(1) rRNA (2) tRNA
(3) hnRNA (4) All of these

121. Which is not true about splicing?
(1) Introns are removed and exons are joined in a defined order.
(2) Exons are removed and introns are joined in a defined order
(3) Presence of non-coding sequence makes the primary transcripts non-functional
(4) Splicing occurs only in eukaryotes

122. Which of the following is not a stop codon?
(1) UGA (2) UAA
(3) UGU (4) UAG

123. Which codon performs dual function?
(1) AGU (2) UGA
(3) AUC (4) AUG

124. $\underset{(i)}{DNA} \xrightarrow{(ii)} mRNA \xrightarrow{(iii)} protein$

Identify (i), (ii) and (ii) and who proposed the given central dogma?
(1) (i) → Translation, (ii) → Replication, (iii) → Transcription → James Watson
(2) (i) → Replication, (ii) → Transcription, (iii) → Translation → Francis crick
(3) (i) → Transcription, (ii) → Replication, (iii) → Translation → Francis crick
(4) (i) → Replication, (ii) → Transcription, (iii) → Transcription → James Watson

125. Which of the following is correct regarding genetic code?
(1) UUU is the initiation codon which also codes for phenylalanie.
(2) There are 64 triplet codons and only 20 amino acids.
(3) Three random nitrogen bases specify the placement of one amino acid.
(4) UAA is the nonsense codon which also codes for methionine.

126. Which is true about ribonucleic acid?
(1) RNA can not generate its replica.
(2) RNA mutate at a very slow rate.
(3) RNA is highly reactive and catalytic.
(4) RNA is most stable due to presence of uracil.

127. Identify the incorrect pair:
(1) *E. coli.* → 4.6×10^6 bp
(2) Diploid human DNA → 6.6×10^9 bp
(3) Bacteriophage λ → 48502 nucleotides
(4) $\phi \times 174$ → 53860 nucleotides

128. Select true (T) or false (F) for the following statements and choose the correct option
A. The experiment of Hershey and chase gave unequivocal proof that DNA is the genetic material.
B. Free 2'OH of RNA makes it more labile and easily degradable.
C. Hershey and chase used bacteria in their experiment to confirm that DNA is the genetic material in organisms.
D. Bacteria infected with bacteriophage radiolabeled with ^{35}P were radioactive

	A	**B**	**C**	**D**
(1)	T	T	F	F
(2)	T	F	T	F
(3)	T	F	F	F
(4)	T	F	T	T

129. DNA replicates semi-conservatively was proved by
(1) Griffith
(2) Meselson and Stahl
(3) Hershey and Chase
(4) Beadle and Tatum

130. Match the following and choose the correct option

(A) Jacob and Monad	(i) Cell free system
(B) Alec Jeffery	(ii) *Neurospora Crassa*
(C) Nirenberg	(iii) Operon
(D) One gene one enzyme	(iv) Viruses
(E) Capsid	(v) DNA fingerprinting

	A	B	C	D	E
(1)	iii	v	i	ii	iv
(2)	i	ii	iii	iv	v
(3)	iii	i	v	iv	ii
(4)	v	i	ii	iii	iv

131. A B-DNA molecule completes one turn after every
(1) Base pair (2) 10 bp
(3) 12 bp (4) 8 bp

132. VNTR size varies from
(1) 0.1-20 kb (2) 1.0-20 kb
(3) 0.2-10 kb (4) 0.1-200 kb

133. **Assertion:** DNA molecule acts as a template for synthesis of both DNA and RNA.
Reason: DNA double helix has feature of right handed coiling which alone confers stability to DNA.
(1) Both assertion and reason are true and reason is correct explanation of assertion.
(2) Both assertion and reason are true and reason is not correct explanation of assertion.
(3) Assertion is true but reason is false
(4) Assertion is false but reason is true

134. Regulation of gene expression occurs at various levels in eukaryotes.
Choose the correct sequence of these levels
(1) Gene → Primary transcript → Splicing → mRNA → Translation → Protein
(2) Primary transcript → Gene → mRNA → Splicing → translation → Protein
(3) Gene → mRNA → Primary transcript → Splicing → Translation → Protein
(4) Gene → Primary transcript → Splicing → mRNA → Splicing → Translation

135. In lac operon, gene y, z, i, a codes for
(1) β-galactosidase, permease, repressor protein, transacetylase
(2) Permease, β-galactosidase, repressor protein, transacetylase
(3) β-galactosidase, transacetylase, permease, repressor protein
(4) Transacetylase, permease, β-galactosidase, repressor protein

SECTION-B

136. The inducer for switching on and off of the lac operon is
(1) Sucrose (2) Repressor
(3) Promoter (4) Lactose

137. Match the codons with their coded amino acids and choose the correct option

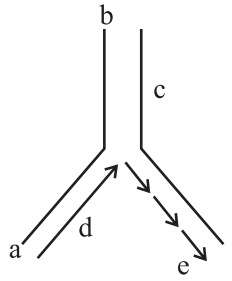
(A) AUG	(i) Cysteine
(B) UUU	(ii) Glycine
(C) UGU	(iii) Methionine
(D) GGG	(iv) Phenylalanine

	A	B	C	D
(1)	iii	iv	ii	i
(2)	i	ii	iii	iv
(3)	iii	ii	iv	i
(4)	iii	iv	i	ii

138. Which is true for translation?
(1) UTRs are present before stop codon.
(2) Translation begins when large subunit of ribosome encounters an mRNA.
(3) Ribozyme catalyse the formation of peptide bond.
(4) For terminating translation, a release factor binds to start and stop codon.

139. Methyl guanosine triphosphate is related to
(1) Tailing (2) Capping
(3) Dual Function (4) Okazaki fragments

140. Mention the polarity of DNA, strand a-b and Identify c, d & c in the following diagram?



(1) a-b →3′ → 5′, c-template strand, d-leading strand, e-lagging strand
(2) a-b →5′ → 3′, c-template strand, d-lagging strand, e- leading strand
(3) a-b →3′ → 5′, c-template strand, d-lagging strand, e- leading strand
(4) All are incorrect

141. **Statement I:** The functions are unknown for over 60% of the discovered genes.
Statement II: Less than 2% of the genome codes for proteins.
(1) Both statements are correct
(2) Both statements are incorrect
(3) Only statement I is correct
(4) Only statement II is correct

142. Polymorphism in DNA sequence forms the basis of:
(1) Genetic mapping
(2) DNA fingerprinting
(3) Mutation
(4) Both (1) and (2)

143. During density gradient centrifugation small peaks formed are referred to as
(1) Satellite DNA (2) Genomic DNA
(3) Bulk DNA (4) Both (1) and (3)

144. DNA fingerprinting involve following steps. Arrange them in correct sequence.
A. Isolation of DNA
B. Separation of DNA by gel electrophoresis
C. Digestion of DNA by endonucleases
D. Blotting
E. Autoradiography
F. Hybridisation using VNTR probe
(1) $B \rightarrow A \rightarrow C \rightarrow D \rightarrow F \rightarrow E$
(2) $C \rightarrow A \rightarrow B \rightarrow D \rightarrow E \rightarrow F$
(3) $A \rightarrow C \rightarrow B \rightarrow D \rightarrow F \rightarrow E$
(4) $A \rightarrow B \rightarrow C \rightarrow D \rightarrow F \rightarrow E$

145. In DNA fingerprinting, what is used as probe
(1) Bulk DNA (2) Genomic DNA
(3) Both (1) and (2) (4) Satellite DNA

146. Identify the correct statements and choose the correct answer.
A. Heterochromatin are darkly stained and transcriptionally active
B. Euchromatin are lightly stained and transcriptionally active
C. Frequency of crossing over is high in euchromatin
D. Frequency of crossing over is low in euchromatin
E. Heterochromatin are mostly affected by temperature and age.
(1) A, B, D are correct
(2) B, C are correct
(3) B, C, E are correct
(4) All are incorrect

147. **Assertion:** Only one strand of DNA is being copied during transcription called template strand.
Reason: Copying of both strand would prevent translation of RNA into protein.
(1) Both assertion and reason are true and reason is correct explanation of assertion.
(2) Both assertion and reason are true and reason is not correct explanation of assertion.
(3) Assertion is true but reason is false
(4) Assertion is false but reason is true

148. **Assertion:** Transcription and translation could be coupled in E. coli.
Reason: There is no well-defined nucleus present in *E. coli.*
(1) Both assertion and reason are true and reason is correct explanation of assertion.
(2) Both assertion and reason are true and reason is not correct explanation of assertion.
(3) Assertion is true but reason is false
(4) Assertion is false but reason is true

149. Which of the following is not a goal of human Gerome project?
(1) To determine sequence of 3 billion chemical base pairs that make human DNA
(2) Store the information of genes & bp in databases.
(3) Elimination of all diseases
(4) Addressing ELSI

150. Okazaki fragments are joined by enzyme
(1) Topoisomerase
(2) Polymerase
(3) Helicase
(4) Ligase

.

SECTION-IV (ZOOLOGY)

SECTION – A

151. Life cannot originate from inorganic materials because of
(1) High degree of environment pollution
(2) Very high atmospheric temperature
(3) Very high amount of oxygen in the atmosphere
(4) Absence of raw materials

152. Match column I with column II and choose the correct option

Column I		Column II	
A.	Saltation	1.	Spores
B.	Darwin	2.	Founder effect
C.	Genetic drift	3.	Evening primose
D.	Panspermia	4.	Finches

	A	B	C	D
(1)	3	4	2	1
(2)	3	2	1	4
(3)	3	4	1	2
(4)	1	4	2	3

153. Water broke up into oxygen and hydrogen in early atmosphere due to
(1) Ultraviolet radiations
(2) X-rays
(3) Gamma rays
(4) Infrared rays

154. Which organism was taken by Pasteur for his experiment?
(1) Yeast (2) *E. coli*
(3) *Salmonella* (4) *Streptococcus*

155. The conditions present on the early earth were:
(1) High temperature
(2) Volcanic eruptions (storms)
(3) Reducing atmosphere containing CH_4, NH_3 etc.
(4) All of these

156. Which gases had taken up by S.L. Miller for his experimentation?
(1) CH_4, H_2 (2) NH_3
(3) H_2O (4) All of these

157. The first cells were supposed to be originated on earth about
(1) 2 billion years ago
(2) 2000 million years ago
(3) 3000 million years ago
(4) Both (1) and (2)

158. The first form of life arose in _____ environment.
(1) Aquatic (2) Terrestrial
(3) Arboreal (4) Aerial

159. According to theory of special creation earth was supposed to be
(1) 4000 years old
(2) 2000 years old
(3) 4 billion years old
(4) 4 million years old

160. Given below are two statements.
Statement I: Louis Pasteur conducted experiments on killed yeast.
Statement II: Louis Pasteur disproved the theory of special creation.
Choose the correct option:
(1) Statement I is correct but statement II is incorrect
(2) Statement I is incorrect but statement II is correct
(3) Both Statement I and II are correct
(4) Both Statement I and II are incorrect

161. What was the term given by Darwin for the differences in characteristics?
(1) Natural selection
(2) Fitness
(3) Evolution
(4) Variation

162. Fossils are dated by
(1) Amount of calcium residue in them
(2) Their sodium content
(3) Their number in an area
(4) Amount of radioactive carbon isotope in them

163. Analogous organs/structures are
(1) Similar in origin
(2) Similar in function
(3) Non-functional
(4) Similar in anatomy

164. The divergent evolution can be seen in
(1) Analogous structures
(2) Artificial breeding
(3) Drug resistance
(4) Homologous structures

165. Fossilisation can occur when
(1) Animals are buried and preserved by natural process
(2) Animals are destroyed by the scavengers
(3) Animals are eaten up by predators
(4) Animals are destroyed by environmental condition

166. Forelimbs of man, cheetah, whale and bat are examples of
(1) Homologous structures
(2) Analogous structures
(3) Convergent evolution
(4) All of these

167. The occurrence of endemic species in South America and Australia is due to
(1) Extinction of these species from other regions
(2) Continental separation
(3) No terrestrial route to these places
(4) Retrogressive evolution

168. Example of divergent evolution is
(1) Thorns of *Bougainvillea* and tendrils of C*ucurbita*
(2) Wings of butterfly and birds
(3) Sweet potato and potato
(4) All of these

169. Analogous structures are result of
(1) Divergent evolution
(2) Convergent evolution
(3) Parallel evolution
(4) Both (1) and (2)

170. Given below are two statements.
Statement I: Flippers of Penguins and Dolphins are example of analogous structures.
Statement II: Analogous structures are the result of divergent evolution.
Choose the correct option.
(1) Statement I is correct but statement II is incorrect
(2) Statement I is incorrect but statement II is correct
(3) Both statement I and II are correct
(4) Both statement I and II are incorrect

171. Examples of natural selection are
(1) Industrial melanism
(2) Antibiotic and drug resistance
(3) Herbicides resistance
(4) All of these

172. Natural selection was given by
(1) Darwin
(2) Louis Pasteur
(3) Lamarck
(4) Hardy

173. Choose the incorrect statement about natural selection.
(1) Before industrialisation tree trunk were darker in colour
(2) Before industrialisation tree trunk were lighter in colour.
(3) Before industrialisation white coloured lichens were present on tree trunk
(4) Before industrialisation there were no deposition of smoke and soot on tree trunk

174. Smoke has _____ that eradicated the lichens.
(1) SO_2 (2) CO_2
(3) NO_2 (4) All of the above

175. Lichens are
(1) Algae
(2) Fungi
(3) Pollution indicator
(4) Animal

176. The paleontological evidences are obtained from:
(1) Homologous structure
(2) Analogous structure
(3) Fossils
(4) Lichens

177. Which of the following was a land dinosaur?
(1) *Triceratops*
(2) *Pteranodon*
(3) *Archaeopteryx*
(4) *Ichthyosaur*

178. Presence of gills in the tadpole of frog indicates that
(1) Fishes were amphibians in the past
(2) Fishes evolved from frog like ancestors
(3) Frogs will have gills in future
(4) Frogs evolved from gills ancestors

179. The Darwinian variations are–
(1) Small and directionless
(2) Random and directional
(3) Small and directional
(4) Random and directionless

180. **Assertion (A):** Inter specific hybrids are usually sterile
Reason (R): Interspecific hybrids receives chromosomes from two different species
(1) Both (A) and (R) are true and the (R) is the correct explanation of (A)
(2) Both (A) and (R) are true but (R) is not the correct explanation of (A)
(3) (A) is true statement but (R) is false
(4) Both (A) and (R) are false statements

181. Adaptive radiation is exemplified by
(1) Darwin's finches
(2) Australian marsupials
(3) Placental mammals
(4) All of these

182. Variation of beaks of finches that Darwin found in Galapagos Island is due to
(1) Head movement
(2) Food eating habits
(3) Eye blinking
(4) Both (1) and (2)

183. Theory of Panspermia was proposed to explain the
(1) Origin of universe
(2) Origin of life
(3) Origin of earth
(4) Evolution of life

184. Evolutionary biology is the study of
(1) Civilisation
(2) History of life forms on earth
(3) Hardy-Weinberg equilibrium
(4) Reproductive success

185. If gene migration happen multiple times, it would lead to
(1) Mutation
(2) Gene flow
(3) Chemical evolution
(4) Genetic drift

SECTION: B

186. First life forms on earth were:
(1) Cyanobacteria
(2) Chemoheterotrophs
(3) Autotrophs
(4) Photoautotrophs

187. Among following, the most recent is
(1) *Homo sapiens*
(2) *Australopithecus*
(3) *Homo habilis*
(4) *Dryopithecus*

188. *Australopithecines* probably lived in the
(1) East Africa grasslands
(2) West Africa grasslands
(3) North Africa grasslands
(4) South Africa grasslands

189. The Neanderthal man had cranial capacity of around
(1) 1600 cc (2) 650-800 cc
(3) 900 cc (4) 1400 cc

190. **Assertion (A):** In 1938, a coelacanth was found in South Africa.
Reason (R): Pelycosaurs and Therapsids were synapsids.
(1) Both (A) and (R) are true and the (R) is the correct explanation of the (A)
(2) Both (A) and (R) are true but (R) is not the correct explanation of (A)
(3) Both (A) and (R) are false
(4) (A) is true but (R) is false

191. The homologous organs are those that show similarity in
(1) Diameter (2) Function
(3) Origin (4) Size

192. All Australian marsupials evolved in the
(1) Africa
(2) Galapagos island
(3) Australian continent
(4) North America

193. Fitness, according to Darwin, refers ultimately and only to
(1) Physically dominant over other
(2) Ability to defend from other
(3) Inheritance of acquired characters
(4) Number of offsprings produce

194. The idea not related to the Darwinian theory of evolution is
(1) Survival of the fittest
(2) Struggle for existence
(3) Inheritance of acquired characters
(4) Origin of species by natural selection

195. Lamarck gave the example of _____ to explain his theory.
(1) Giraffes (2) Finches
(3) Marsupials (4) Placental mammals

196. The factor/s affect hardy-Weinberg equilibrium is are :
(1) Genetic drift
(2) Gene migration
(3) Mutation
(4) All of these

197. Genetic drift operates in
(1) Small isolated population
(2) Large isolated population
(3) Non-reproductive population
(4) Slow reproductive population

198. Saltation means
(1) Single step small mutation
(2) Single step small variation
(3) Single step large mutation
(4) Single step large variation

199. If nature selects both of the peripheral traits of a characters than which kind of natural selection to be operating?
(1) Stabilising (2) Disruptive
(3) Directional (4) Rotating

200. **Assertion (A):** Both mule and hinny are sterile
Reason (R): These are the example of hybrid sterility
Chose the correct option
(1) Both (A) and (R) are true and the (R) is the correct explanation of (A)
(2) Both (A) and (R) are true and the (R) is not the correct explanation of (A)
(3) (A) is true and (R) is false
(4) Both (A) and (R) are false



PW Web/App - https://smart.link/7wwosivoicgd4

Library- https://smart.link/sdfez8ejd80if